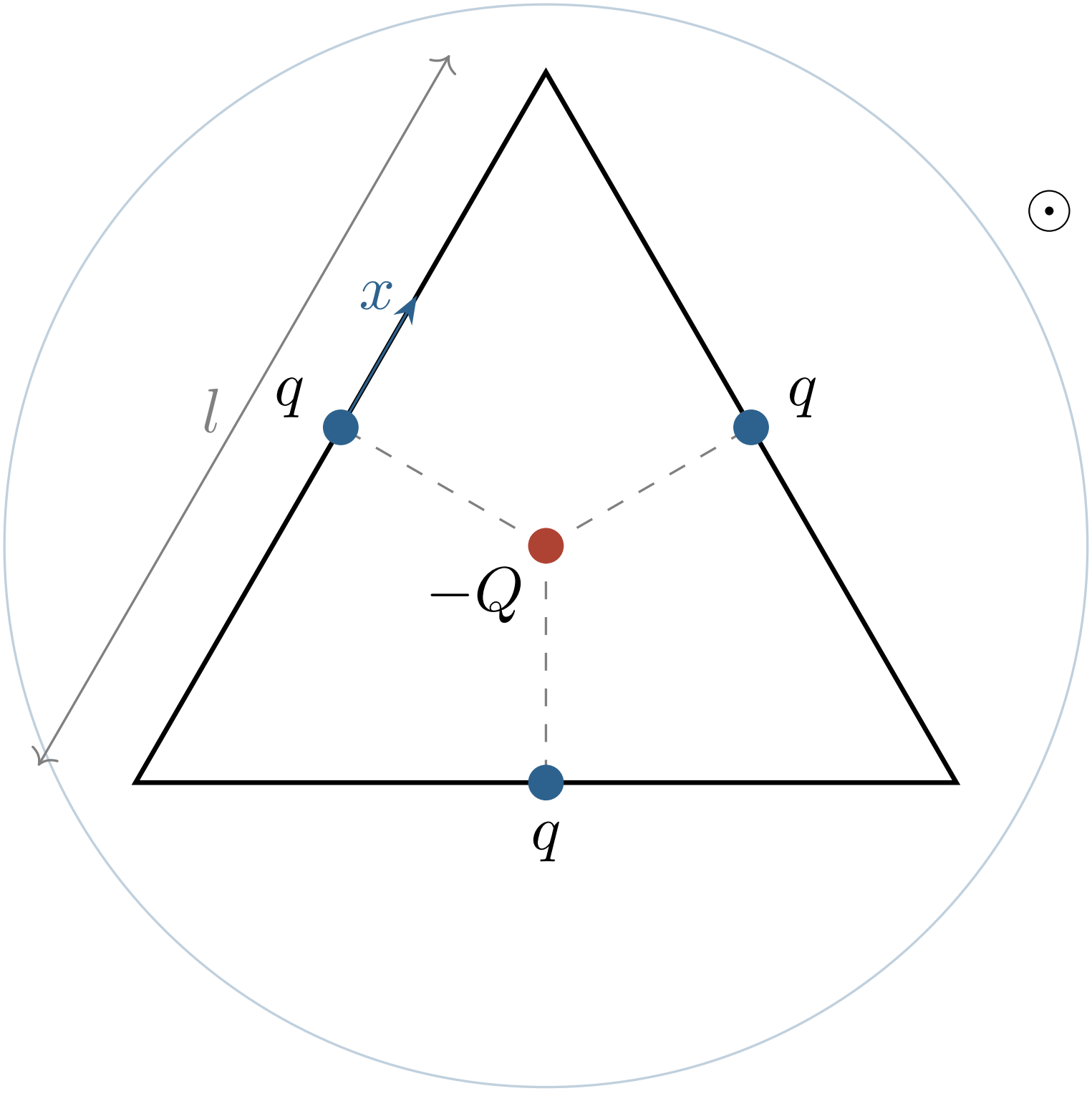
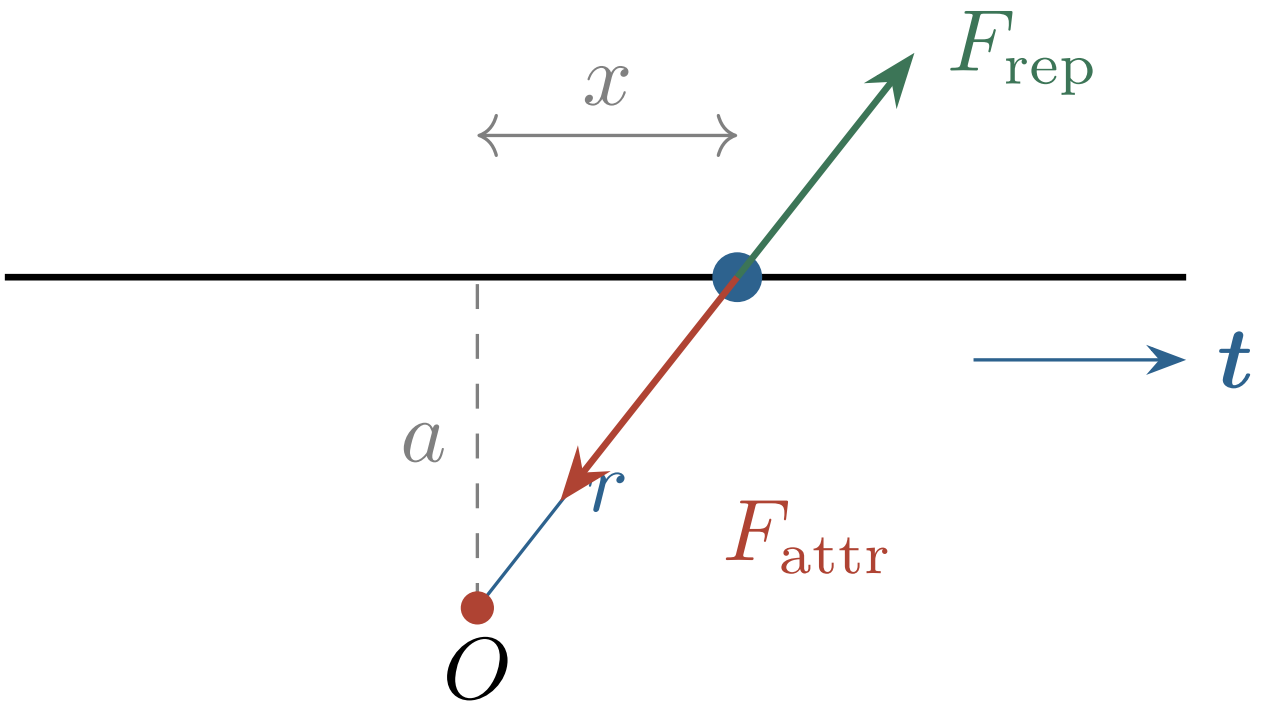
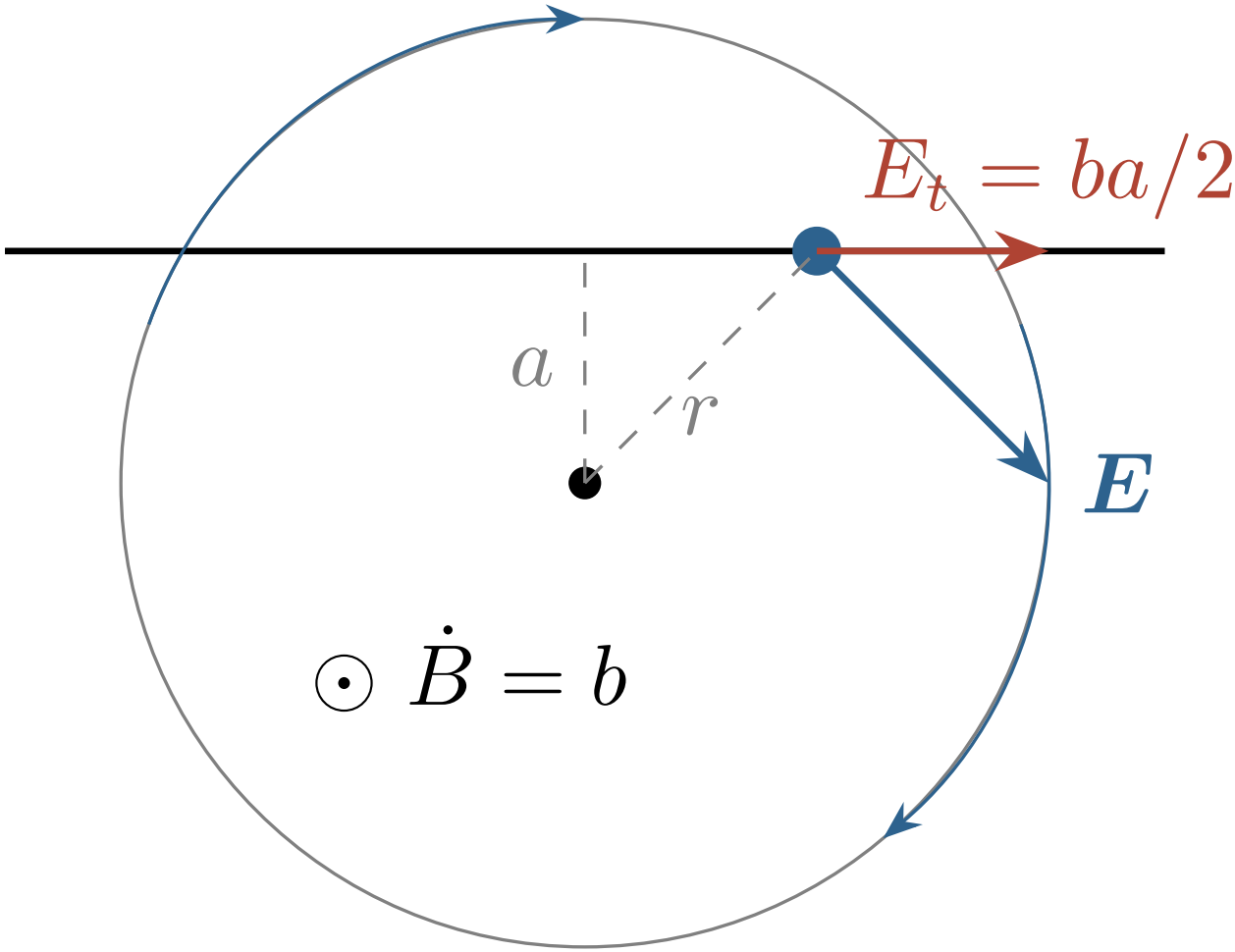


俯视图；绳沿三条槽边闭合

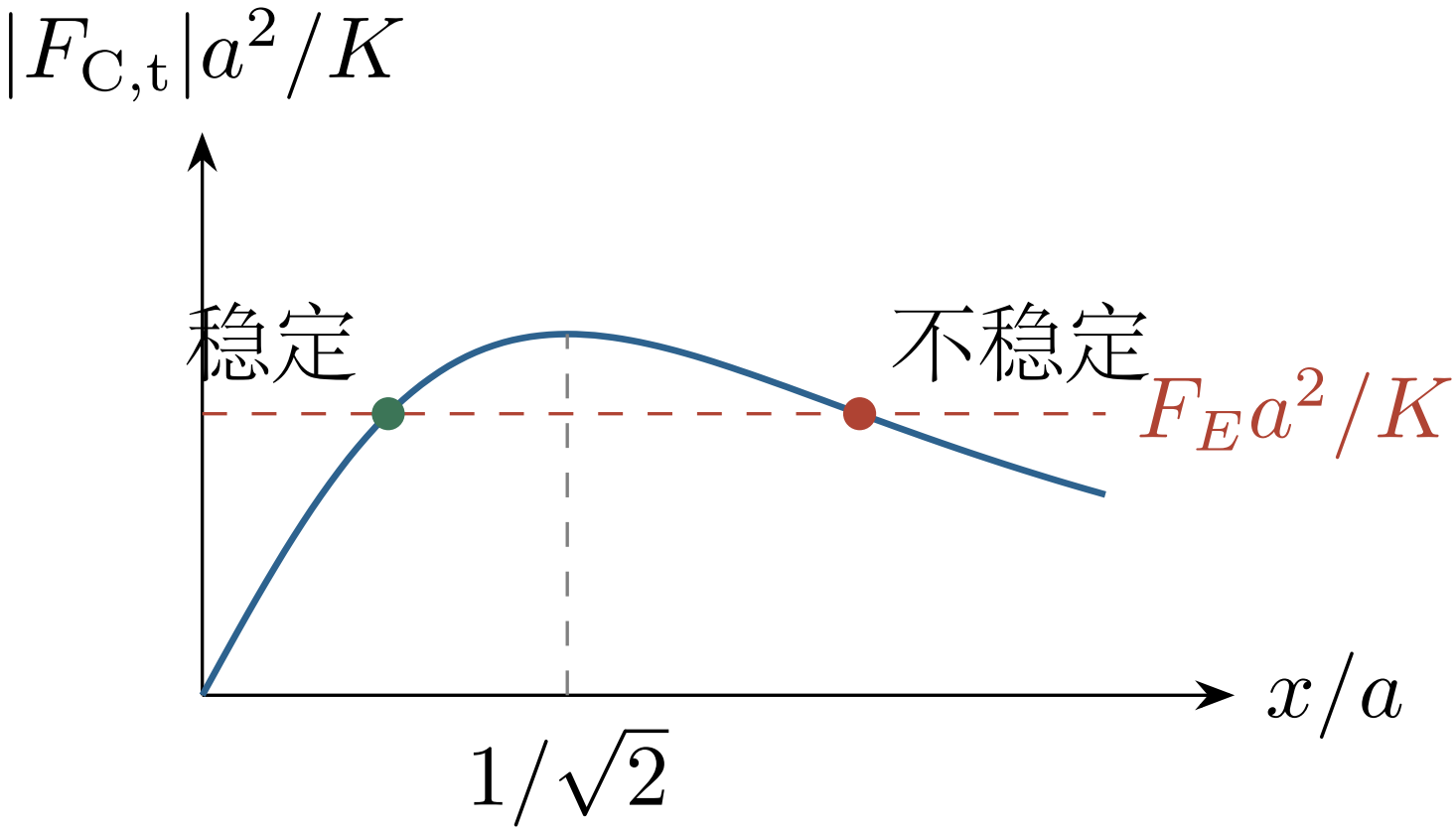


$$F_{\mathrm{rep}} = k_e q^2/(\sqrt{3}r^2)$$

$$F_{\mathrm{C,t}} = -Kx/r^3$$



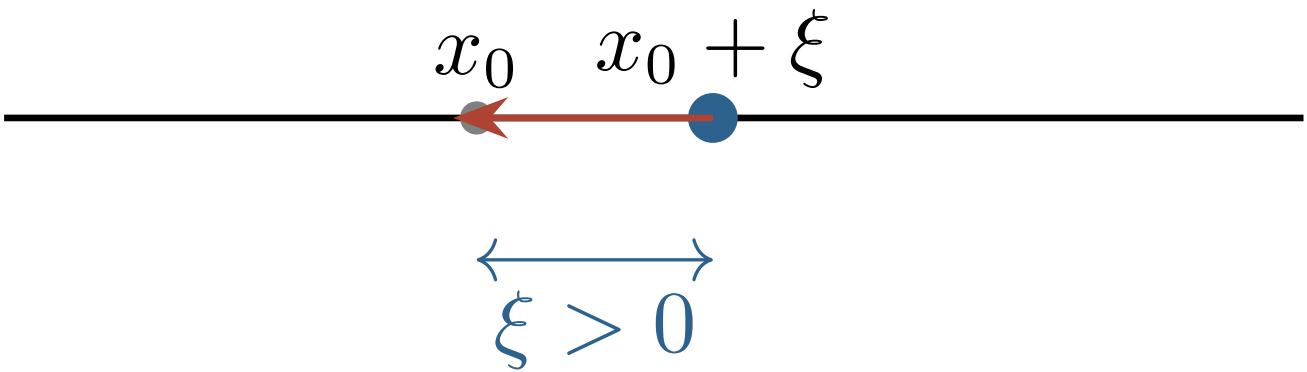
沿槽顺时针为正；E_t 与 x 无关



增长支：恢复作用增强

衰减支：位移受正反馈

$$F \simeq F'(x_0)\xi < 0$$



$$m\ddot{\xi} + \kappa\xi = 0$$

$$\kappa = K(a^2 - 2x_0^2)/(a^2 + x_0^2)^{5/2} > 0$$